Document Control

Title **Intranasal Fentanyl for Treatment of Children with acute pain in ED Guidelines**			
Author			**Author's job title** Associate Specialist in Emergency Medicine
Directorate			**Department** Emergency Department
Version	**Date Issued**	**Status**	**Comment / Changes / Approval**
0.1	July 2019	Draft	Initial version for consultation/ ED Governance
1.0	Nov 2019	Final	Approved at Clinical Audit and Guidelines Group 12th November 2019.
Main Contact AS in Emergency Medicine North Devon District Hospital Raleigh Park Barnstaple, EX31 4JB			**Tel: Direct Dial** – **Tel: Internal** – **Email**:
Lead Director Medical Director			
Superseded Documents			
Issue Date Nov 2019	**Review Date** Nov 2022	**Review Cycle** Three years	
Consulted with the following stakeholders: (list all) • Consultants in Emergency Department • Lead Pharmacist Women's and Children			
Approval and Review Process • Lead Clinician Emergency Department • Pharmacy • ED Governance (14th August 2019) • Clinical Audit and Guidelines Group			
Local Archive Reference **Local Path** **Filename** Policy Template (NHSLA) v1.16 20Jul11.doc			
Policy categories for Trust's internal website (Bob) Pharmacy/ ED Guidelines			**Tags for Trust's internal website (Bob)** INF for pain relief in children IN Fentanyl in Children

CONTENTS

Document Control 1
1. Purpose 3
2. Definitions 3
3. Responsibilities 3
4. IN Fentanyl 4
5 Monitoring Compliance with and the Effectiveness of the Guideline 6
6 References 6

1. Purpose

1.1. The purpose of this document is to detail the process for use of Intranasal Fentanyl in children with acute moderate-severe pain in ED

1.2. The policy applies to ED staff

1.3. Implementation of this policy will ensure that:

- Acute pain in children is managed in timely manner in ED
- All ED staff are familiar with the dose calculation and application of IN fentanyl in children

2. Definitions

Definition

2.1. IN= Intra nasal

2.2. ED= Emergency Department

2.3. The goal of this guideline is to provide timely management of acute pain in children in ED as it is much easier and quicker to prepare and administer compared to IN diamorphine and IV morphine

3. Responsibilities

3.1 Responsibility of the clinician introducing the guideline

- Ensuring that all staff are aware of the fentanyl availability, location
- Ensuring that staff are familiar with dose calculation and administration of the drug in children
- Ensuring that the pharmacy lead is happy to support and stock the fentanyl in ED
- Audit the new procedure and presenting to the local clinical governance group

3.2 Responsibility of Emergency Department Staff

- Checking the drug on a daily basis using the medicines policy guidelines
- Safe storage of the drug
- Safe administration of the drug as per the guideline
- Correct recording of the administration
- Safely monitoring the patient post administration
- Ordering stocks of the drug according to the medicines policy
- Maintaining competency to administer the drug safely

4. IN Fentanyl

4.1 Mechanism of Action

Fentanyl gets absorbed through the vascular nasal mucosa, avoiding the first pass metabolism and breakdown by the liver, hence the quick onset of action. Therapeutic levels are reached within 2 minutes and may last for up to 30 minutes. The dose can be repeated if adequate analgesia is not achieved after the first dose.

4.2 Indications

- Children >1 year with moderate-severe pain
- Splint application
- Wound cleaning
- Fracture manipulation
- Burns
- Foreign body removal

4.3 Dose

Intranasal Fentanyl Dosage Schedule (using 100 mcg per 2 ml)

For weights that fall between ranges, please give the smaller dose.
Eg. 13 kg will receive the dose for 10-12 kg, which is 0.3 ml and 0.1 ml top up dose

Intranasal fentanyl Dosage schedule (using 100mcg per 2 ml)

For weights that fall between ranges: Please give the smaller dose

E.g. 13kg will receive the dose for the 10-12 kg, which is 0.3 ml and 0.1 ml top up dose.

estimated weight (kg)	initial dose (1.5mcg/kg)	volume - initial dose (ml)	top up dose (0.5 mcg/kg) (To be given 30minutes after the initial dose if required)	volume - top up dose (ml)
7 - 8	10 mcg	0.2 ml	3.5 - 4 mcg	0 07 ml
10 - 12	15 mcg	0.3 ml		0.1 ml
14 - 16	20 mcg	0.4 ml	7 - 8 mcg	0.15 ml
18 - 20	27 mcg	0.55 ml	9 - 10 mcg	0.18 ml
22 - 24	33 mcg	0. 65 ml	11 - 12 mcg	0.22 ml
26 - 28	39 mcg	0.8 ml	13 - 14 mcg	0.26 ml
30 - 32	45 mcg	0.9 ml	15 - 16 mcg	0.3 ml
34 - 38	51 mcg	1.0 ml	17 - 19 mcg	0.35 ml
40 - 44	60 mcg	1.2 ml	20 - 22 mcg	0.4 ml
46 - 48	69 mcg	1.4 ml	23 - 24 mcg	0.45 ml
> 50	75 mcg	1.5 ml	25 mcg	0.5 ml

4.4 Side Effects

- Bad taste
- Vomiting (very rare)
- Rare serious systemic side effects if given in correct dose, which includes hypotension and respiratory depression.

4.5 Caution

- Symptoms of URTI and nasal congestion might cause unreliable delivery of the drug
- Epistaxis
- Prior dosing with opioids

4.6 Relative Contraindications

- Age< 1 year (limited data on safety or efficacy in this age group)
- Head injury
- Hypovolaemia

4.7 Equipment Needed

- 100mcg/2ml Fentanyl
- 1ml Syringe
- Mucosal atomiser device

4.8 Technique

- The patient should be reclining at 45 degrees. The syringe should be held horizontal and the contents expelled into the nares in one rapid dose (Do not ask the patient to sniff).
- Doses of 1 ml (50mcg) or more should be divided between nares

4.9 Observation, monitoring and recovery

- Routine observation (O2 saturation, Respiratory Rate, BP, HR every 5 minutes for the next 20 minutes
- Can be discharged home after 1 hour
- Routine post sedation advice should be given to the parents/ carers on discharge

5 Monitoring Compliance with and the Effectiveness of the Guideline

Standards/ Key Performance Indicators

Key performance indicators comprise:

- Feedback from staff based on ease of preparation and administration
- Patient's pain score before and after administration

Process for Implementation and Monitoring Compliance and Effectiveness

Replace diamorphine stock with fentanyl

Inform all staff about the new changes during morning/afternoon handover

Send email to all ED staff informing them of the new guidelines

Posters around the ED with all the information about the new drug

Poster on the drug cupboard explaining the dose, preparation and administration

To monitor compliance we can audit it based on the pain score and staff satisfaction after 3 months of implementation or for the first 20 patients(whichever comes first)

All doctors and staff have the responsibility of following the guidelines but the introducing clinician will be responsible for monitoring and auditing compliance after 3 months. Non-compliance will be reported to the responsible line manager and will be discussed in the Clinical Governance meeting to consolidate staff compliance.

6 References

1.Murphy A, O'Sullivan R, Wakai A, Grant TS, Barrett MJ, Cronin J, McCoy SC, Hom J, Kandamany N.
Intranasal fentanyl for the management of acute pain in children.
Cochrane Database of Systematic Reviews 2014, Issue 10. Art. No.: CD009942.
DOI: 10.1002/14651858.CD009942.pub2.

2. A Randomized Controlled Trial Comparing Intranasal Fentanyl to Intravenous Morphine for Managing Acute Pain in Children
in the Emergency Department
Meredith Borland, MBBS,FACEM,Ian Jacobs, PhD, FRCNA,Barbara King, MBBS, FRACP,Debra O'Brien, MBBS, FACEM

3. Starship.org.nz
4. https://www.northdevonhealth.nhs.uk/2016/10/controlled-drugs-policy/